

Unit 3. Multiple Regression II: Inference and Interpretation

Throughout, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ under the assumptions A1–A5 of Unit 2, with $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$, $s^2 = \text{SSE}/(n - k - 1)$, and $\mathbf{H}$, $\mathbf{M}$ as defined there.

1 Inference for a Single Coefficient

FREES §3.4 · ISLR §3.2.2 · CLM §7.1, §7.4.

Theorem 3.1 (*t* test for one coefficient). Under A1–A5, for $H_0: \beta_j = \beta_j^0$,

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\text{SE}(\hat{\beta}_j)} \sim t_{n-k-1} \quad \text{under } H_0, \quad \text{SE}(\hat{\beta}_j) = s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}. \quad (1)$$

Proof. Step 1. By the sampling distribution of $\hat{\boldsymbol{\beta}}$ established in Unit 2, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$; the j -th coordinate of a multivariate normal is normal with the (j, j) variance, so

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim N(0, 1).$$

Step 2. $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ independently of $\hat{\boldsymbol{\beta}}$, hence of Z .

Step 3. $Z/\sqrt{V/(n-k-1)}$ is t_{n-k-1} by definition, and

$$\frac{Z}{\sqrt{V/(n-k-1)}} = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\text{MSE}} \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{\text{SE}(\hat{\beta}_j)}.$$

□

Theorem 3.2 (Confidence interval for $\mathbf{c}'\boldsymbol{\beta}$). Under A1–A5, a $(1 - \alpha)$ confidence interval for $\mathbf{c}'\boldsymbol{\beta}$ is

$$\mathbf{c}'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} s \sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}. \quad (2)$$

Proof. $\mathbf{c}'\hat{\boldsymbol{\beta}} \sim N(\mathbf{c}'\boldsymbol{\beta}, \sigma^2 \mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$ by closure of the normal under linear maps. Standardising and dividing by $\sqrt{V/(n-k-1)}$ as in Theorem 3.1 gives

$$\frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s \sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim t_{n-k-1},$$

and inverting the pivot yields (2). □

Remark. $\hat{\beta}_j$ is the case $\mathbf{c} = \mathbf{e}_j$. The interpretation of β_j is the change in $E[Y]$ per unit change in x_j with the other explanatory variables held fixed.

2 Inference for Several Coefficients

FREES §3.4 · ISLR §3.2.2 · CLM §7.3.

Theorem 3.3 (General linear hypothesis). Let $\mathbf{R}$ be $q \times (k + 1)$ of rank q and $\mathbf{r} \in \mathbb{R}^q$. Under A1–A5, for $H_0: \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$,

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q s^2} \sim F_{q, n-k-1} \quad \text{under } H_0. \quad (3)$$

Proof. Step 1. $\mathbf{R}\hat{\boldsymbol{\beta}} \sim N(\mathbf{R}\boldsymbol{\beta}, \sigma^2 \boldsymbol{\Sigma}_R)$ with $\boldsymbol{\Sigma}_R = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$, so under H_0 , $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r} \sim N(\mathbf{0}, \sigma^2 \boldsymbol{\Sigma}_R)$.

Step 2. $\boldsymbol{\Sigma}_R$ is positive definite: for $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}'\boldsymbol{\Sigma}_R\mathbf{v} = (\mathbf{R}'\mathbf{v})'(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{R}'\mathbf{v}) > 0$, since $\text{rank}(\mathbf{R}) = q$ makes $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$. With $\boldsymbol{\Sigma}_R^{1/2}$ its symmetric positive definite square root and $\mathbf{w} = \sigma^{-1}\boldsymbol{\Sigma}_R^{-1/2}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$

$$\mathbf{r}) \sim N(\mathbf{0}, \mathbf{I}_q),$$

$$W = \mathbf{w}'\mathbf{w} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{\sigma^2} \sim \chi_q^2.$$

Step 3. W is a function of $\hat{\boldsymbol{\beta}}$, hence independent of $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$. Then $F = \{W/q\}/\{V/(n-k-1)\}$ is $F_{q,n-k-1}$, and substituting $s^2 = \text{SSE}/(n-k-1)$ gives (3). $\square$

Proposition 3.4 (Restricted least squares). Under A4, with $\mathbf{R}$ of rank q , the minimiser of $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ is

$$\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}), \quad (4)$$

and the residual sums of squares satisfy

$$\text{SSE}_R - \text{SSE}_U = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}). \quad (5)$$

Proof. Write $\mathbf{A} = \mathbf{X}'\mathbf{X}$, which is positive definite by A4, and $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}} + \mathbf{d}$. Since $\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$,

the cross term vanishes and

$$\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 - 2\mathbf{d}'\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \mathbf{d}'\mathbf{A}\mathbf{d} = \text{SSE}_U + \mathbf{d}'\mathbf{A}\mathbf{d}.$$

The constraint $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ becomes $\mathbf{R}\mathbf{d} = \mathbf{c}$ with $\mathbf{c} = \mathbf{r} - \mathbf{R}\hat{\boldsymbol{\beta}}$, so the problem is to minimise $\mathbf{d}'\mathbf{A}\mathbf{d}$ over that affine set. Put $\mathbf{d}^* = \mathbf{A}^{-1}\mathbf{R}'\boldsymbol{\mu}$ with $\boldsymbol{\mu} = (\mathbf{R}\mathbf{A}^{-1}\mathbf{R}')^{-1}\mathbf{c}$, which is defined because $\mathbf{R}\mathbf{A}^{-1}\mathbf{R}'$ is positive definite (Step 2 of Theorem 3.3) and which satisfies $\mathbf{R}\mathbf{d}^* = \mathbf{c}$. Any competitor is $\mathbf{d}^* + \mathbf{v}$ with $\mathbf{R}\mathbf{v} = \mathbf{0}$, and

$$(\mathbf{d}^* + \mathbf{v})'\mathbf{A}(\mathbf{d}^* + \mathbf{v}) = \mathbf{d}^{*'}\mathbf{A}\mathbf{d}^* + 2\boldsymbol{\mu}'\mathbf{R}\mathbf{v} + \mathbf{v}'\mathbf{A}\mathbf{v} = \mathbf{d}^{*'}\mathbf{A}\mathbf{d}^* + \mathbf{v}'\mathbf{A}\mathbf{v},$$

using $\mathbf{d}^{*'}\mathbf{A}\mathbf{v} = \boldsymbol{\mu}'\mathbf{R}\mathbf{A}^{-1}\mathbf{A}\mathbf{v} = \boldsymbol{\mu}'\mathbf{R}\mathbf{v} = 0$. The last term is non-negative and vanishes only at $\mathbf{v} = \mathbf{0}$, so $\mathbf{d}^*$ is the unique minimiser, which is (4). Its value is

$$\mathbf{d}^{*'}\mathbf{A}\mathbf{d}^* = \boldsymbol{\mu}'\mathbf{R}\mathbf{A}^{-1}\mathbf{R}'\boldsymbol{\mu} = \mathbf{c}'(\mathbf{R}\mathbf{A}^{-1}\mathbf{R}')^{-1}\mathbf{c},$$

and $\text{SSE}_R = \text{SSE}_U + \mathbf{d}^{*'}\mathbf{A}\mathbf{d}^*$ gives (5), the sign of $\mathbf{c}$ being immaterial in a quadratic form. $\square$

Corollary 3.5 (Extra sum of squares). Let SSE_U be the residual sum of squares of the full model and SSE_R that of the model fitted under the q constraints $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$. Then

$$F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n - k - 1)}. \quad (6)$$

Proof. By (5), $\text{SSE}_R - \text{SSE}_U$ is exactly the numerator quadratic form of (3). Dividing by $q s^2$ turns (6) into (3) term by term, so the two are the same number for every data set, not merely two statistics with the same law, and Theorem 3.3 gives the $F_{q,n-k-1}$ distribution under H_0 . $\square$

Corollary 3.6 ($t^2 = F$ for one constraint). For $H_0: \mathbf{c}'\boldsymbol{\beta} = r_0$ ($q = 1$), the statistic (3) is the square of

$$t = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}}.$$

Proof. With $\mathbf{R} = \mathbf{c}'$, $\mathbf{r} = r_0$, $\Sigma_R = \mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}$ is a scalar, so

$$F = \frac{(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)^2}{s^2 \mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}} = t^2. \quad \square$$

Remark (Which $\mathbf{R}$ gives which test). The overall F of Unit 2 is $\mathbf{R} = [\mathbf{0} \mid \mathbf{I}_k]$, $\mathbf{r} = \mathbf{0}$, so $q = k$. Dropping a subset of q predictors is $\mathbf{R}$ selecting those rows, and (6) is then the **partial** F test: fit with and without them and compare residual sums of squares. Testing $\beta_1 = \beta_2$ is $\mathbf{R} = (0, 1, -1, 0, \dots)$ with $\mathbf{r} = 0$, a constraint that is not a variable exclusion, and for which (3) is the only convenient route.

3 Prediction

FREES §3.4 ·CLM §9.

Theorem 3.7 (Confidence interval for the mean; prediction interval). Let $\mathbf{x}_0 \in \mathbb{R}^{k+1}$ and $\hat{Y}_0 =$

$\mathbf{x}'_0 \hat{\boldsymbol{\beta}}$. Under A1–A5, a $(1 - \alpha)$ confidence interval for $E[Y_0 \mid \mathbf{x}_0] = \mathbf{x}'_0 \boldsymbol{\beta}$ is

$$\mathbf{x}'_0 \hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} s \sqrt{\mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0}, \quad (7)$$

and a $(1 - \alpha)$ prediction interval for a new $Y_0 = \mathbf{x}'_0 \boldsymbol{\beta} + \varepsilon_0$ is

$$\mathbf{x}'_0 \hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} s \sqrt{1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0}. \quad (8)$$

Proof. (7) is Theorem 3.2 with $\mathbf{c} = \mathbf{x}_0$. For (8), the prediction error is $\hat{Y}_0 - Y_0 = \mathbf{x}'_0 (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \varepsilon_0$. The new error ε_0 is independent of the sample errors $\boldsymbol{\varepsilon}$, and $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ is a function of $\boldsymbol{\varepsilon}$, so the two pieces are independent and

$$\text{var}(\hat{Y}_0 - Y_0) = \mathbf{x}'_0 \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{x}_0 + \sigma^2 = \sigma^2 (1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0).$$

The error is normal as a linear combination of normals, and independent of SSE, so replacing σ by s gives a t_{n-k-1} pivot. $\square$

Corollary 3.8 (The case $k = 1$). With a single explanatory variable, $\mathbf{x}_0 = (1, x_0)'$ and $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{(n-1)s_x^2}$, so the two intervals are

$$\hat{y}_0 \pm t_{\alpha/2, n-2} s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{(n-1)s_x^2}}, \quad \hat{y}_0 \pm t_{\alpha/2, n-2} s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{(n-1)s_x^2}}.$$

Proof. Using the explicit inverse of $\mathbf{X}'\mathbf{X}$ computed in Unit 2 for $\mathbf{X} = [\mathbf{1} \ \mathbf{x}]$,

$$\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \frac{\sum_i x_i^2 - 2n\bar{x}x_0 + nx_0^2}{n \sum_i (x_i - \bar{x})^2} = \frac{\sum_i (x_i - \bar{x})^2 + n(x_0 - \bar{x})^2}{n \sum_i (x_i - \bar{x})^2},$$

the numerator following from $\sum_i x_i^2 = \sum_i (x_i - \bar{x})^2 + n\bar{x}^2$ and $n\bar{x}^2 - 2n\bar{x}x_0 + nx_0^2 = n(x_0 - \bar{x})^2$. Splitting the fraction gives $1/n + (x_0 - \bar{x})^2/\{(n-1)s_x^2\}$. $\square$

Remark. Both widths grow quadratically as x_0 leaves $\bar{x}$, since $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$ is the leverage the new point would have: extrapolation is where the model is least supported and the interval widest.

The two differ only by the $1+$, which does not vanish as $n \rightarrow \infty$: the mean response becomes known exactly, a single future observation never does.

4 Special Explanatory Variables

FREES §3.5 · ISLR §3.3.1–3.3.2.

4.1 Binary variables

Definition 3.9 (Indicator variable). A **binary** or **indicator** variable takes the values 0 and 1, coding the absence or presence of an attribute.

Remark. In $\mathbf{E}[y] = \beta_0 + \beta_1 x + \beta_2 z$ with z binary, the fitted relation is two parallel lines: intercept β_0 when $z = 0$ and $\beta_0 + \beta_2$ when $z = 1$, common slope β_1 . So β_2 is the vertical distance between the two groups at every x , and testing $H_0: \beta_2 = 0$ tests whether the groups differ once x is accounted for.

4.2 Categorical variables with several levels

Definition 3.10 (Reference coding). A categorical variable with c levels is represented by $c - 1$ indicators, one level being omitted as the **reference**. Each coefficient is the difference between its level and the reference, at fixed values of the other explanatory variables.

Proposition 3.11 (Why $c - 1$ and not c). If all c indicators are included alongside an intercept, $\mathbf{X}$ has rank at most k , so A4 fails and $\hat{\boldsymbol{\beta}}$ is not unique.

Proof. The c indicator columns sum to $\mathbf{1}$ at every observation, since each observation belongs to exactly one level. That is a linear dependence among the columns of $\mathbf{X}$ including the intercept column, so $\text{rank}(\mathbf{X}) < k + 1$ and $\mathbf{X}'\mathbf{X}$ is singular by Step 3 of the rank argument in the OLS theorem of Unit 2. $\square$

Remark. Software resolves the singularity silently: R drops the redundant column and reports NA for its coefficient. Changing the reference level changes every reported coefficient and every individual t statistic, but not the fitted values, SSE, R^2 , or the F test for the factor as a whole. This is why a categorical variable should be assessed by the partial F of Corollary 3.5 on all $c - 1$

indicators jointly, not by individual t statistics.

4.3 Interactions

Definition 3.12 (Interaction). An **interaction** term is a product of two explanatory variables entered as a further column of $\mathbf{X}$.

Remark. In $\mathbf{E}[y] = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 xz$ with z binary, the slope in x is β_1 when $z = 0$ and $\beta_1 + \beta_3$ when $z = 1$: the two lines are no longer parallel, and β_3 is the difference in slopes. When x and z are both continuous, $\partial \mathbf{E}[y]/\partial x = \beta_1 + \beta_3 z$ depends on z , so no single number describes the effect of x . A main effect is then not interpretable in isolation, which is the reason for the **hierarchy principle**: if xz is in the model, keep x and z too, whatever their individual t statistics.

4.4 Transformed variables

Remark (Reading transformed coefficients). The model stays linear in the parameters, so all of §§1–3 applies unchanged; only interpretation changes.

- $E[y] = \beta_0 + \beta_1 \ln x$: replacing x by $1.01x$ adds $\beta_1 \ln(1.01) = 0.00995\beta_1$, so a 1% increase in x changes $E[y]$ by about $\beta_1/100$ units.
- $E[\ln y] = \beta_0 + \beta_1 x$: a one-unit increase in x adds β_1 to $E[\ln y]$, multiplying the fitted value of y by e^{β_1} , a change of $100(e^{\beta_1} - 1)\%$, which is about $100\beta_1\%$ for small β_1 .
- $E[\ln y] = \beta_0 + \beta_1 \ln x$: $dE[\ln y]/d \ln x = \beta_1$, so β_1 is an elasticity, the percentage change in y per percentage change in x .
- $E[y] = \beta_0 + \beta_1 x + \beta_2 x^2$: differentiating, the effect of x is $\beta_1 + 2\beta_2 x$, and the model is still linear in $(\beta_0, \beta_1, \beta_2)$.

The first two are first-order approximations whose error grows with $|\beta_1|$; the third and fourth are exact.

5 Interpretation

FREES §6.1 ·CLM §8.

5.1 The Frisch–Waugh–Lovell theorem

Theorem 3.13 (Frisch–Waugh–Lovell). Partition $\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2]$ and $\boldsymbol{\beta} = (\boldsymbol{\beta}'_1, \boldsymbol{\beta}'_2)'$, and let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2$. Then

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{M}_2\mathbf{Y}, \quad (9)$$

and the residuals of the full regression equal those of regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$.

Proof. The normal equations for the partitioned model are

$$\mathbf{X}'_1\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}'_1\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}'_1\mathbf{Y}, \quad \mathbf{X}'_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}'_2\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}'_2\mathbf{Y}.$$

Solving the second for $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1)$ and substituting into the first,

$$\mathbf{X}'_1\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}'_1\mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{X}'_1\mathbf{Y}.$$

Writing $\mathbf{H}_2 = \mathbf{I} - \mathbf{M}_2$ and collecting terms in $\hat{\beta}_1$,

$$\mathbf{X}'_1(\mathbf{I} - \mathbf{H}_2)\mathbf{X}_1\hat{\beta}_1 = \mathbf{X}'_1(\mathbf{I} - \mathbf{H}_2)\mathbf{Y},$$

which is $\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1\hat{\beta}_1 = \mathbf{X}'_1\mathbf{M}_2\mathbf{Y}$, giving (9). Since $\mathbf{M}_2$ is symmetric idempotent, $\mathbf{X}'_1\mathbf{M}_2\mathbf{Y} = (\mathbf{M}_2\mathbf{X}_1)'(\mathbf{M}_2\mathbf{Y})$ and $\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1 = (\mathbf{M}_2\mathbf{X}_1)'(\mathbf{M}_2\mathbf{X}_1)$, so (9) is exactly the least squares coefficient from regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$; the residual claim follows because both regressions produce $\mathbf{MY}$. $\square$

Remark. This is what “holding the other variables fixed” means operationally: $\hat{\beta}_j$ is the slope from regressing the part of $\mathbf{Y}$ unexplained by the other predictors on the part of $\mathbf{x}_j$ unexplained by them. No observation need have the other variables actually held fixed; the adjustment is arithmetic, and it is only as good as the variables available to adjust for.

5.2 Omitted-variable bias

Theorem 3.14 (Omitted-variable bias). Suppose $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ but only $\mathbf{X}_1$ is fitted, giving $\tilde{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y}$. Then

$$\mathbb{E}[\tilde{\boldsymbol{\beta}}_1] = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2. \quad (10)$$

Proof. Substituting the true $\mathbf{Y}$,

$$\tilde{\boldsymbol{\beta}}_1 = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\boldsymbol{\varepsilon},$$

and the last term has expectation $\mathbf{0}$ by [A2](#). □

Remark. The bias is the coefficient matrix from regressing the omitted $\mathbf{X}_2$ on the included $\mathbf{X}_1$, multiplied by the omitted effect $\boldsymbol{\beta}_2$. It vanishes iff $\boldsymbol{\beta}_2 = \mathbf{0}$ or $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$. With one included and one omitted variable the sign of the bias is the product of the sign of $\boldsymbol{\beta}_2$ and the sign of the sample correlation between the two variables, which is enough to reason about the direction of the error without knowing its size.

Remark (Significance, size and causation). Significance is not importance: $\text{SE}(\hat{\beta}_j) \propto n^{-1/2}$, so at fixed effect size $|t|$ grows without bound in n , and with small n a large effect can fail to reach significance. Nor does significance establish causation from observational data, since Theorem 3.14 shows that any unmeasured variable related to both response and predictor displaces the coefficient. A coefficient is a comparison, adjusted for what was measured.

6 Computation: Formulas Against Package Output

ISLR §3.6.3, §3.6.6.

```
1 library(ISLR2)
2 fit <- lm(medv ~ lstat + age + rm, data = Boston)
3 X <- model.matrix(fit); y <- Boston$medv
4 n <- nrow(X); k <- ncol(X) - 1L
5 b <- coef(fit); XtXi <- solve(t(X) %*% X)
6 s2 <- sum(residuals(fit)^2) / (n - k - 1)
7 c(n = n, k = k, df = n - k - 1)
```

```
##      n      k      df
## 506      3      502
```

6.1 One coefficient (§1)

```

1 se <- sqrt(s2 * diag(XtXi))           # s sqrt([(X'X)^{-1}]_{jj})
2 t_m <- b / se
3 ci <- cbind(b - qt(0.975, n-k-1) * se, b + qt(0.975, n-k-1) * se)
4 tab <- coef(summary(fit))
5 round(cbind(se, se_lm = tab[, 2], t_m, t_lm = tab[, 3],
6            ci, confint(fit)), 6)

```

```

##              se    se_lm      t_m      t_lm
## (Intercept) 3.18192 3.18192  -0.3694  -0.3694 -7.42684
## lstat       0.05436 0.05436 -12.2985 -12.2985 -0.77531
## age        0.01121 0.01121   0.8106   0.8106 -0.01294
## rm         0.45431 0.45431  11.0479  11.0479  4.12656
##
##              2.5 %   97.5 %
## (Intercept)  5.07622 -7.42684  5.07622
## lstat       -0.56172 -0.77531 -0.56172
## age         0.03112 -0.01294  0.03112
## rm          5.91171  4.12656  5.91171

```

A linear combination: the fitted difference in **medv** between two properties differing by +1 in **rm** and

-2 in `lstat`, i.e. $\mathbf{c}'\boldsymbol{\beta}$ with $\mathbf{c} = (0, -2, 0, 1)'$.

```
1 cc <- c(0, -2, 0, 1)
2 est <- sum(cc * b); se_c <- sqrt(s2 * as.numeric(t(cc) %*% XtXi %*% cc))
3 c(estimate = est, se = se_c,
4   lower = est - qt(0.975, n-k-1)*se_c, upper = est + qt(0.975, n-k-1)*se_c,
5   t = est / se_c)
```

```
## estimate      se      lower      upper      t
##      6.356    0.398    5.574    7.138   15.970
```

6.2 Several coefficients (§2)

$H_0: \beta_{\text{age}} = \beta_{\text{rm}} = 0$, by four routes: the quadratic form (3), the extra sum of squares (6), `anova` on the nested pair, and `car::linearHypothesis`.

```
1 R <- rbind(c(0, 0, 1, 0), c(0, 0, 0, 1)); r0 <- c(0, 0); q <- nrow(R)
2 d <- R %*% b - r0
3 F_quad <- as.numeric(t(d) %*% solve(R %*% XtXi %*% t(R)) %*% d / (q * s2))
4
5 red <- lm(medv ~ lstat, data = Boston)
6 F_extra <- ((sum(residuals(red)^2) - sum(residuals(fit)^2)) / q) / s2
7
```

```

8 suppressMessages(library(car))
9 c(quadratic_form = F_quad, extra_SS = F_extra,
10   anova           = anova(red, fit)$F[2],
11   linearHyp       = linearHypothesis(fit, c("age = 0", "rm = 0"))$F[2],
12   p_value         = pf(F_quad, q, n-k-1, lower.tail = FALSE))

```

```

## quadratic_form      extra_SS      anova      linearHyp
##      6.598e+01      6.598e+01      6.598e+01      6.598e+01
##      p_value
##      3.621e-26

```

A constraint that is not a variable exclusion, $H_0: \beta_{\text{lstat}} = \beta_{\text{age}}$, for which only the quadratic form is convenient:

```

1 R2m <- matrix(c(0, 1, -1, 0), nrow = 1)
2 d2  <- R2m %*% b
3 F2  <- as.numeric(t(d2) %*% solve(R2m %*% XtXi %*% t(R2m)) %*% d2 / (1 * s2))
4 c(F_quad = F2, linearHyp = linearHypothesis(fit, "lstat = age"))$F[2],
5   t_squared = (as.numeric(d2) / sqrt(s2 * as.numeric(R2m %*% XtXi %*% t(R2m))))^2)

```

```

##      F_quad linearHyp t_squared
##      120.7      120.7      120.7

```

The last two columns agree, which is Corollary 3.6 with $q = 1$.

6.3 Prediction (§3)

```
1 x0 <- c(1, 10, 60, 6.5)
2 nd <- data.frame(lstat = 10, age = 60, rm = 6.5)
3 lev <- as.numeric(t(x0) %*% XtXi %*% x0)      # leverage the new point would have
4 tq  <- qt(0.975, n - k - 1); yh <- sum(x0 * b)
5 rbind(manual = c(yh, yh + c(-1, 1) * tq * sqrt(s2 * lev)),
6        predict = predict(fit, nd, interval = "confidence"))
```

```
##           fit    lwr    upr
## manual 25.31 24.79 25.83
## 1      25.31 24.79 25.83
```

```
1 rbind(manual = c(yh, yh + c(-1, 1) * tq * sqrt(s2 * (1 + lev))),
2        predict = predict(fit, nd, interval = "prediction"))
```

```
##           fit    lwr    upr
## manual 25.31 14.41 36.21
## 1      25.31 14.41 36.21
```

6.4 Categorical variables and interactions (§4)

chas is binary and **rad** is coded as an integer but is really an index of accessibility; treating it as a factor gives $c - 1$ indicators.

```
1 Boston$rad_f <- factor(Boston$rad)
2 m_fac <- lm(medv ~ lstat + rad_f, data = Boston)
3 c(levels = nlevels(Boston$rad_f),
4   columns_of_X = ncol(model.matrix(m_fac)),
5   coefficients = length(coef(m_fac)))
```

```
##          levels columns_of_X coefficients
##              9             10             10
```

Adding all c indicators alongside the intercept makes **X** singular (Proposition 3.11); R drops the redundant column and reports NA.

```
1 D <- model.matrix(~ rad_f - 1, data = Boston) # all c indicators
2 c(rank_with_intercept = qr(cbind(1, D))$rank,
3   columns              = ncol(cbind(1, D)),
4   row_sums_all_one     = all(rowSums(D) == 1))
```

```
## rank_with_intercept          columns    row_sums_all_one
```

```
##                9                10                1
```

```
1 trap <- lm(Boston$medv ~ D)
2 sum(is.na(coef(trap)))           # one coefficient is inestimable
```

```
## [1] 1
```

The factor is assessed by the partial F on all its indicators jointly, not by individual t statistics:

```
1 m_small <- lm(medv ~ lstat, data = Boston)
2 a <- anova(m_small, m_fac)
3 c(F = a$F[2], df1 = a$Df[2], df2 = a$Res.Df[2],
4    p = a$`Pr(>F)`[2])
```

```
##                F                df1                df2                p
## 4.59958282      8.00000000 496.00000000 0.00001985
```

Changing the reference level moves every coefficient but no fitted value:

```
1 m_alt <- lm(medv ~ lstat + relevel(rad_f, ref = "24"), data = Boston)
2 c(max_coef_change = max(abs(coef(m_fac)[-(1:2)] - coef(m_alt)[-(1:2)])),
3   max_fitted_diff = max(abs(fitted(m_fac) - fitted(m_alt))),
4   SSE_same        = sum(residuals(m_fac)^2) - sum(residuals(m_alt)^2))
```

```
## max_coef_change max_fitted_diff      SSE_same
##           6.939e+00      1.421e-14      0.000e+00
```

An interaction makes the slope depend on the other variable:

```
1 m_int <- lm(medv ~ lstat * chas, data = Boston)
2 cf <- coef(m_int)
3 c(slope_chas0 = cf[["lstat"]],                                # beta_1
4   slope_chas1 = cf[["lstat"]] + cf[["lstat:chas"]],          # beta_1 + beta_3
5   difference   = cf[["lstat:chas"]],
6   p_interact   = coef(summary(m_int))["lstat:chas", 4])
```

```
## slope_chas0 slope_chas1 difference p_interact
##    -0.914984   -1.347863   -0.432878    0.007112
```

6.5 FWL and omitted-variable bias (§5)

Regressing out **age** and **rm** from both **medv** and **lstat** reproduces the full-model coefficient on **lstat**, which is Theorem 3.13.

```
1 X2 <- X[, c(1, 3, 4)]                                # intercept, age, rm
2 M2 <- diag(n) - X2 %*% solve(t(X2) %*% X2) %*% t(X2)
3 y_res <- drop(M2 %*% y); x_res <- drop(M2 %*% X[, "lstat"])
```

```

4 c(fwl_lm      = coef(lm(y_res ~ x_res - 1))["x_res"]),
5   fwl_formula = sum(x_res * y_res) / sum(x_res^2),
6   full_model  = b["lstat"],
7   resid_matches = max(abs(residuals(lm(y_res ~ x_res - 1)) - residuals(fit))))

```

```

##           fwl_lm    fwl_formula    full_model resid_matches
## -6.685e-01    -6.685e-01    -6.685e-01     9.308e-13

```

Theorem 3.14 predicts the bias from omitting **rm**: the coefficient from regressing **rm** on the retained variables, times β_{rm} .

```

1 X1 <- X[, c(1, 2, 3)]                                # intercept, lstat, age
2 short <- lm(medv ~ lstat + age, data = Boston)
3 delta <- solve(t(X1) %*% X1, t(X1) %*% X[, "rm"])    # rm regressed on X1
4 predicted_short <- coef(fit)[c(1, 2, 3)] + drop(delta) * b["rm"]
5 round(cbind(short_fit = coef(short), predicted = predicted_short,
6             diff = coef(short) - predicted_short), 10)

```

```

##           short_fit predicted diff
## (Intercept)  33.22276  33.22276    0
## lstat        -1.03207  -1.03207    0
## age           0.03454   0.03454    0

```

The identity is algebraic and holds in the realised sample, not merely in expectation. A simulation with a known truth separates bias from sampling noise:

```

1 set.seed(1)
2 B <- 3000; ns <- 120
3 Z <- X[1:ns, c(1, 2, 3)]; rm_col <- X[1:ns, "rm"]
4 beta1_true <- c(20, -0.6, 0.03); beta_rm_true <- 4
5 delta_s <- solve(t(Z) %*% Z, t(Z) %*% rm_col)
6 est <- t(replicate(B, {
7   ysim <- drop(Z %*% beta1_true) + rm_col * beta_rm_true + rnorm(ns, sd = 4)
8   drop(solve(t(Z) %*% Z, t(Z) %*% ysim))
9 }))
10 round(rbind(truth      = beta1_true,
11             mean_short = colMeans(est),
12             predicted   = beta1_true + drop(delta_s) * beta_rm_true), 4)

```

```

##           (Intercept)   lstat    age
## truth           20.00 -0.6000 0.0300
## mean_short      46.58 -0.9044 0.0583
## predicted       46.58 -0.9059 0.0586

```

7 Pitfalls

7.1 Reading a coefficient as a causal effect

By Theorem 3.14, a coefficient absorbs the effect of any omitted variable correlated with its predictor. Below, $\mathbf{x}$ has no effect on $\mathbf{y}$, but both depend on an unmeasured $\mathbf{u}$.

```
1 set.seed(4)
2 u <- rnorm(400); xx <- u + rnorm(400, sd = 0.4); yy <- 2 * u + rnorm(400, sd = 0.4)
3 c(coef_without_u = coef(lm(yy ~ xx))["xx"],      # large and significant
4   t_without_u     = coef(summary(lm(yy ~ xx)))["xx", "t value"],
5   coef_with_u     = coef(lm(yy ~ xx + u))["xx"])
```

```
## coef_without_u    t_without_u    coef_with_u
##           1.76117         43.69971         0.05199
```

Diagnostic. None available from the fit alone: the omitted variable is by definition not in the data.

Fix. Subject-matter reasoning about what was not measured; a designed study where one is possible.

7.2 Dropping a main effect while keeping the interaction

If xz is in the model, β_1 is the slope of x **at** $z = 0$, not an average effect, so its t statistic answers a question about one point of the z range.

```

1 set.seed(5)
2 z <- rbinom(300, 1, 0.5); x2 <- rnorm(300, mean = 5)
3 y2 <- 1 + 0.5*x2 + 2*z - 0.5*x2*z + rnorm(300, sd = 0.5)
4 full <- lm(y2 ~ x2 * z); drop_main <- lm(y2 ~ x2:z + z)
5 c(full_slope_z0 = coef(full)[["x2"]],
6   full_slope_z1 = coef(full)[["x2"]] + coef(full)[["x2:z"]],
7   dropped_R2    = summary(drop_main)$r.squared,
8   full_R2       = summary(full)$r.squared)

```

```

## full_slope_z0 full_slope_z1    dropped_R2      full_R2
##      0.4994682    -0.0002798      0.1856843      0.4518499

```

Diagnostic. A non-significant main effect alongside a significant interaction. **Fix.** Keep the hierarchy; centre the predictors if the value $z = 0$ is not meaningful, which makes the main effect the slope at the mean.

7.3 Assessing a factor by individual t statistics

Each t compares one level with the reference; none tests the factor.

```

1 tt <- coef(summary(m_fac))[-(1:2), 4] # p-values of the rad indicators
2 c(n_indicators = length(tt), n_p_below_05 = sum(tt < 0.05),
3   min_p = min(tt), joint_F_p = anova(m_small, m_fac)$`Pr(>F)`[2])

```

```

## n_indicators n_p_below_05      min_p    joint_F_p
##      8.000000000      4.000000000  0.00040198  0.00001985

```

Diagnostic. Individually weak indicators with a strong joint F , or the reverse. **Fix.** The partial F of Corollary 3.5 on all $c - 1$ indicators.

8 Exercises

Exercise 3.1. Show that for a single linear constraint the F statistic (3) equals t^2 , and explain why the overall F of Unit 2 is not the square of any individual t when $k > 1$.

Solution. The first part is Corollary 3.6: with $q = 1$ the matrix $\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is a scalar and the quadratic form collapses to a square.

The overall F tests $q = k$ constraints at once. Its numerator is a χ_k^2 quadratic form in all k

slope estimates jointly, which is not the square of any single normal unless $k = 1$. Two predictors can each have $|t| < 2$ while their joint F is large, when their estimates are strongly negatively correlated: each individual t moves one coefficient with the other free, along a direction the data cannot resolve, while the F asks about both at once. ■

Exercise 3.2. Using Proposition 3.4, show that (a) $\tilde{\beta}$ satisfies $\mathbf{R}\tilde{\beta} = \mathbf{r}$; (b) $\tilde{\beta}$ is unbiased when the constraint is true and biased by $-(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r})$ when it is not; (c) $\text{var}(\hat{\beta}) - \text{var}(\tilde{\beta})$ is positive semi-definite. Verify (a) and (c) numerically, and say what (b) and (c) together mean for constraint-then-estimate.

Solution. Write $\mathbf{A} = \mathbf{X}'\mathbf{X}$, $\mathbf{G} = \mathbf{A}^{-1}\mathbf{R}'(\mathbf{R}\mathbf{A}^{-1}\mathbf{R}')^{-1}$ and $\mathbf{P} = \mathbf{G}\mathbf{R}$, so that $\tilde{\beta} = (\mathbf{I} - \mathbf{P})\hat{\beta} + \mathbf{G}\mathbf{r}$.
 (a) $\mathbf{R}\mathbf{G} = \mathbf{R}\mathbf{A}^{-1}\mathbf{R}'(\mathbf{R}\mathbf{A}^{-1}\mathbf{R}')^{-1} = \mathbf{I}_q$, so $\mathbf{R}\tilde{\beta} = \mathbf{R}\hat{\beta} - (\mathbf{R}\hat{\beta} - \mathbf{r}) = \mathbf{r}$.
 (b) $\mathbf{E}[\tilde{\beta}] = \beta - \mathbf{G}(\mathbf{R}\beta - \mathbf{r})$, which is β iff $\mathbf{R}\beta = \mathbf{r}$. Note $\mathbf{P}$ is idempotent, $\mathbf{P}^2 = \mathbf{G}\mathbf{R}\mathbf{G}\mathbf{R} = \mathbf{P}$, though not symmetric.
 (c) $\tilde{\beta} - \mathbf{E}[\tilde{\beta}] = (\mathbf{I} - \mathbf{P})(\hat{\beta} - \beta)$, so by the variance-under-a-linear-map lemma of Unit 2, $\text{var}(\tilde{\beta}) =$

$\sigma^2(\mathbf{I} - \mathbf{P})\mathbf{A}^{-1}(\mathbf{I} - \mathbf{P})'$. Since $\mathbf{P}\mathbf{A}^{-1}\mathbf{P}' = \mathbf{G}(\mathbf{R}\mathbf{A}^{-1}\mathbf{R}')\mathbf{G}' = \mathbf{P}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{P}'$,

$$\text{var}(\hat{\boldsymbol{\beta}}) - \text{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2(\mathbf{P}\mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{P}' - \mathbf{P}\mathbf{A}^{-1}\mathbf{P}') = \sigma^2\mathbf{P}\mathbf{A}^{-1}\mathbf{P}' = \sigma^2\mathbf{G}(\mathbf{R}\mathbf{A}^{-1}\mathbf{R}')\mathbf{G}',$$

positive semi-definite.

```
1 Rc <- rbind(c(0,0,1,0), c(0,0,0,1)); rc <- c(0, 0)
2 Gm <- XtXi %*% t(Rc) %*% solve(Rc %*% XtXi %*% t(Rc))
3 btil <- drop(b - Gm %*% (Rc %*% b - rc))
4 Vres <- s2 * (diag(4) - Gm %*% Rc) %*% XtXi %*% t(diag(4) - Gm %*% Rc)
5 rbind(restricted = round(btil, 6),
6       lm_on_lstat = round(c(coef(red), 0, 0), 6))
```

```
##           (Intercept) lstat age rm
## restricted      34.55 -0.95  0  0
## lm_on_lstat    34.55 -0.95  0  0
```

```
1 rbind(se_unrestricted = round(sqrt(diag(s2*XtXi)), 5),
2       se_restricted   = round(sqrt(diag(Vres)), 5))
```

```
##           (Intercept)  lstat    age    rm
## se_unrestricted    3.1819 0.05436 0.01122 0.4543
## se_restricted      0.5017 0.03454 0.00000 0.0000
```

```

1 c(constraint_holds = max(abs(Rc %*% btil - rc)),
2   se_from_reduced_fit = coef(summary(red))[, 2])

```

```

##               constraint_holds
##               1.041e-17
## se_from_reduced_fit.(Intercept)
##               5.626e-01
##      se_from_reduced_fit.lstat
##               3.873e-02

```

Imposing $\beta_{\text{age}} = \beta_{\text{rm}} = 0$ reproduces `lm(medv ~ lstat)` exactly and cuts the intercept's standard error by a factor of six.

A false constraint buys a variance reduction and pays a bias; the bias does not shrink with n while the variance does, so at large n a wrongly imposed restriction dominates the error. And a constraint retained because its test did not reject is not a constraint known to be true. ■

Exercise 3.3. A full model with $k = 6$ predictors on $n = 60$ observations has $\text{SSE}_U = 420$; dropping three gives $\text{SSE}_R = 610$. Test the three jointly, and find the SSE_R that would leave them at the 5% boundary.

Solution. By (6) with $q = 3$ and $n - k - 1 = 53$,

$$F = \frac{(610 - 420)/3}{420/53} = \frac{63.33}{7.925} = 7.99$$

on $F_{3,53}$, against a 5% critical value of 2.78: the three are jointly significant.

At the boundary, $\text{SSE}_R - \text{SSE}_U = 3(2.78)(7.925) = 66.1$, so $\text{SSE}_R = 486.1$. Anything the three explain beyond 66 of the reduced model's 610, about 11%, clears the threshold; the observed 190 is nearly three times that. ■

Exercise 3.4. Write $\mathbf{R}$ and $\mathbf{r}$ for each hypothesis in a model with intercept and predictors x_1, x_2, x_3 : (a) $\beta_2 = 0$; (b) $\beta_1 = \beta_2 = \beta_3 = 0$; (c) $\beta_1 = \beta_2$; (d) $\beta_1 + \beta_2 = 1$ and $\beta_3 = 0$. For (c) and (d) give the reduced model explicitly and check its parameter count.

Solution. With $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)'$:

(a) $\mathbf{R} = (0, 0, 1, 0)$, $\mathbf{r} = 0$, $q = 1$.

(b) $\mathbf{R} = [\mathbf{0}_{3 \times 1} \mid \mathbf{I}_3]$, $\mathbf{r} = \mathbf{0}_3$, $q = 3$: the overall F .

(c) $\mathbf{R} = (0, 1, -1, 0)$, $\mathbf{r} = 0$, $q = 1$. Not a variable exclusion. Under $\beta_1 = \beta_2 = \gamma$ the mean is $\beta_0 + \gamma(x_1 + x_2) + \beta_3 x_3$, so the reduced model regresses y on the constructed variable $x_1 + x_2$ and on x_3 : three parameters against four, which is $k + 1 - q$.

(d) $\mathbf{R} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, $\mathbf{r} = (1, 0)'$, $q = 2$. Substituting $\beta_2 = 1 - \beta_1$ and $\beta_3 = 0$ gives mean $\beta_0 + \beta_1(x_1 - x_2) + x_2$, so the reduced model regresses $y - x_2$ on $x_1 - x_2$: two parameters, again $k + 1 - q$. A non-zero $\mathbf{r}$ always appears as an offset on the response. ■

Exercise 3.5. Simulate $n = 60$ observations with two predictors of sample correlation about 0.996 and $y = 1 + v_1 + v_2 + \varepsilon$, $\varepsilon \sim N(0, 2^2)$. Report the two t statistics and the joint F , and explain from $\text{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

Solution.

```
1 set.seed(31)
2 v1 <- rnorm(60); v2 <- 0.995*v1 + sqrt(1 - 0.995^2)*rnorm(60)
3 yv <- 1 + v1 + v2 + rnorm(60, sd = 2)
4 mc <- lm(yv ~ v1 + v2)
5 round(coef(summary(mc)), 4)
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.9278      0.2435  3.8101  0.0003
## v1           3.8544      2.6469  1.4562  0.1508
## v2          -1.7769      2.7064 -0.6566  0.5141
```

```
1 c(cor_predictors = cor(v1, v2), joint_F = summary(mc)$fstatistic[["value"]],
2   p_value = pf(summary(mc)$fstatistic[["value"]], 2, 57, lower.tail = FALSE),
3   cor_of_estimates = cov2cor(vcov(mc))[2, 3])
```

```
## cor_predictors      joint_F      p_value
##      9.956e-01      3.701e+01      4.989e-11
## cor_of_estimates
##      -9.956e-01
```

Neither coefficient is significant at 5% and one has the wrong sign, yet the joint F is 37.0 with $p = 5 \times 10^{-11}$.

The correlation between $\hat{\beta}_1$ and $\hat{\beta}_2$ is exactly minus the correlation between the predictors, which is what $(\mathbf{X}'\mathbf{X})^{-1}$ does to a two-column design after the intercept is removed. The data determine $\beta_1 + \beta_2$ sharply, since $v_1 + v_2$ is essentially $2v_1$, and $\beta_1 - \beta_2$ hardly at all. Each individual t asks about one coefficient with the other free, along the direction the data cannot resolve; the joint F

asks about both and sees the sharp direction. A set of individually insignificant predictors with a significant joint F is not a set of useless predictors. ■

Exercise 3.6. Two individual 95% intervals are not a 95% region for the pair. (a) Give the exact 95% region implied by Theorem 3.3. (b) State the Bonferroni rectangle and its guarantee. (c) By simulation at $n = 60$, measure the coverage of the rectangle of individual intervals and of the F region at predictor correlations $\rho = 0, 0.9, 0.99$, and explain why the rectangle's coverage **rises** with ρ .

Solution. (a) Taking $\mathbf{R}$ to select the two coefficients and inverting the acceptance region of (3), the set of $\mathbf{r}$ not rejected at level α is

$$\{\mathbf{r} : (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \leq 2s^2 F_{1-\alpha; 2, n-k-1}\},$$

an ellipse centred at $\mathbf{R}\hat{\boldsymbol{\beta}}$ with axes along the eigenvectors of $\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$. Its coverage is exactly $1 - \alpha$ by construction.

(b) Bonferroni takes each interval at level $1 - \alpha/2$, so $\mathbf{P}\{\text{either misses}\} \leq \alpha/2 + \alpha/2 = \alpha$. The guarantee is one sided: coverage is **at least** $1 - \alpha$, usually more.

```

1 covr <- function(rho, Rr = 20000, nn = 60) {
2   set.seed(32)
3   z1 <- rnorm(nn); z2 <- rho*z1 + sqrt(1 - rho^2)*rnorm(nn)
4   Xd <- cbind(1, z1, z2); Ad <- solve(crossprod(Xd)); bt <- c(0, 1, 1)
5   tq2 <- qt(0.975, nn - 3); fq <- qf(0.95, 2, nn - 3)
6   rect <- ell <- logical(Rr)
7   for (rr in seq_len(Rr)) {
8     m <- lm.fit(Xd, drop(Xd %*% bt) + rnorm(nn))
9     s2d <- sum(m$residuals^2)/(nn - 3); dh <- m$coefficients[2:3] - bt[2:3]
10    rect[rr] <- all(abs(dh) <= tq2*sqrt(s2d*diag(Ad)[2:3]))
11    ell[rr] <- drop(t(dh) %*% solve(Ad[2:3, 2:3]) %*% dh)/(2*s2d) <= fq
12  }
13  c(rho = rho, rectangle = mean(rect), F_region = mean(ell))
14 }
15 t(sapply(c(0, 0.9, 0.99), covr))

```

```

##      rho rectangle F_region
## [1,] 0.00      0.9060  0.9526
## [2,] 0.90      0.9306  0.9526
## [3,] 0.99      0.9436  0.9526

```

(c) The F region is at nominal coverage throughout. The rectangle is not: at $\rho = 0$ the two events are nearly independent, so joint coverage is about $0.95^2 = 0.9025$.

The F column is bit-identical across the three rows, and not by accident. Since $z_2 = \rho z_1 + \sqrt{1 - \rho^2} w$ with the same z_1 and w , the three designs have the **same column space**, so by the invariance of a projection to the basis chosen for it the numerator $\|(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}\|^2$ and the denominator $\|\mathbf{M}\mathbf{Y}\|^2$ are unchanged and F is literally the same number in every replicate. Only the individual coefficients, which are coordinates in a basis, move.

Coverage rises with ρ because the two events become positively dependent as the estimates become correlated: when the design nearly cannot separate β_1 from β_2 the two estimates err together, so they tend to miss together and the probability that at least one misses falls. The rectangle is therefore anti-conservative by up to five points, worst when the predictors are well separated. ■

Exercise 3.7. At $\mathbf{x}_0$ the leverage is $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 0.04$, with $s = 5$, $n = 120$, $k = 3$. Give the half-widths of the 95% confidence and prediction intervals, and the leverage at which the prediction interval is twice the confidence interval.

Solution. $df = 116$ and $t_{0.975,116} = 1.981$. By (7), $1.981(5)\sqrt{0.04} = 1.98$; by (8), $1.981(5)\sqrt{1.04} = 10.10$, about five times wider: at this leverage the new observation's own noise dominates.

The ratio is $\sqrt{(1+h)/h}$, equal to 2 when $h = 1/3$. Since the mean leverage of an observed point is $(k+1)/n$, a ratio as small as 2 needs h eighty times the average at $n = 120$. For any point resembling the data the prediction interval is many times the confidence interval. ■

Exercise 3.8. For `medv` on `lstat` in `Boston`, compute the 95% confidence and prediction half-widths at $x_0 = \bar{x}$, $\bar{x} + 2s_x$ and $\bar{x} + 10s_x$. Which interval is misleading at the last point, and why is that not visible from the arithmetic?

Solution.

```
1 slr <- lm(medv ~ lstat, data = Boston)
2 xb <- mean(Boston$lstat); sx <- sd(Boston$lstat)
3 Sxx <- sum((Boston$lstat - xb)^2); ss <- summary(slr)$sigma
4 tq3 <- qt(0.975, n - 2)
5 t(sapply(c(0, 2, 10), function(mm) {
6   x0v <- xb + mm*sx; lv <- 1/n + (x0v - xb)^2/Sxx
```

```

7  c(sd_from_mean = mm, x0 = x0v, leverage = lv,
8    CI_half = tq3*ss*sqrt(lv), PI_half = tq3*ss*sqrt(1 + lv),
9    ratio = sqrt((1 + lv)/lv))
10 }))

```

```

##      sd_from_mean      x0 leverage CI_half PI_half ratio
## [1,]           0 12.65 0.001976  0.5429  12.22 22.52
## [2,]           2 26.94 0.009897  1.2149  12.27 10.10
## [3,]          10 84.06 0.199996  5.4613  13.38  2.45

```

```

1  c(max_lstat_observed = max(Boston$lstat))

```

```

## max_lstat_observed
##           37.97

```

The confidence half-width grows tenfold across the table while the prediction half-width grows by under 10%: away from the data the estimation error grows, but σ^2 dominates the prediction interval throughout, so it looks reassuringly stable.

That is what makes it misleading. At $\bar{x} + 10s_x$ we have $x_0 = 84.1$, while **lstat** is a percentage whose observed maximum is 37.97: the point is outside the data and outside the variable's range. Both formulas assume the linear model holds at x_0 , and no quantity computed from the sample

can test that, because there are no observations there. Extrapolation is a modelling failure, not a variance problem. ■

Exercise 3.9. Let (t, u) be the 95% confidence interval and (v, w) the 95% prediction interval at the same $\mathbf{x}_0$. (a) Show $u - t \rightarrow 0$ as $n \rightarrow \infty$ with the predictors sampled from a fixed distribution, and give the exact rate. (b) Show $w - v$ does **not** tend to 0, and give its limit. (c) Show $w - v > u - t$ always, and give the condition under which the two coincide.

Solution. (a) The half-width is $t_{\alpha/2, n-k-1} s \sqrt{\mathbf{x}_0' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0}$. With the rows of $\mathbf{X}$ sampled from a fixed law, $n^{-1} \mathbf{X}'\mathbf{X} \rightarrow \mathbf{\Sigma}$ positive definite, so $\mathbf{x}_0' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0 = n^{-1} \mathbf{x}_0' \mathbf{\Sigma}^{-1} \mathbf{x}_0 \{1 + o(1)\} = O(1/n)$. Also $s \rightarrow \sigma$ and $t_{\alpha/2, n-k-1} \rightarrow z_{\alpha/2}$, so

$$u - t = \frac{2z_{\alpha/2} \sigma \sqrt{\mathbf{x}_0' \mathbf{\Sigma}^{-1} \mathbf{x}_0}}{\sqrt{n}} \{1 + o(1)\} \rightarrow 0,$$

at rate $n^{-1/2}$.

(b) The prediction half-width is $t_{\alpha/2, n-k-1} s \sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$ and the 1 does not shrink, so

$$w - v \longrightarrow 2z_{\alpha/2} \sigma,$$

which is 3.92σ at $\alpha = 0.05$: the interval one would use knowing $\boldsymbol{\beta}$ and σ^2 exactly. Perfect knowledge of the model leaves the new observation's own noise.

(c) Comparing (7) and (8), the two share the factor $t_{\alpha/2, n-k-1} s$ and differ only in $\sqrt{h}$ against $\sqrt{1+h}$, and $\sqrt{1+h} > \sqrt{h}$ for every $h > 0$. So $w - v > u - t$ strictly whenever $s > 0$. The two coincide only in the degenerate case $s = 0$, that is $\text{SSE} = 0$: with no residual variation both intervals collapse to the point $\hat{Y}_0$, which is then exact. Unit 2's perfectly collinear three-point regression is such a case.

The confidence interval is for the fixed number $\mathbf{x}_0'\boldsymbol{\beta}$; the prediction interval is for the random Y_0 . Quoting the first when the second is wanted understates the uncertainty by the factor $\sqrt{(1+h)/h}$, which Exercise 3.8 shows can exceed 20. ■

Exercise 3.10 (SOA sample question 62). A simple linear regression on more than 100 observations reports slope -1.03 with standard error 0.06 . Calculate the 95% confidence interval for the slope. Then give the t statistic and its p -value, the sample size at which the normal and t multipliers differ by less than 1%, and the n needed to halve the interval.

Solution. By Theorem 3.1 the interval is $b_1 \pm t_{0.975, n-2} \text{SE}(b_1)$. With $n > 100$, $t_{0.975, n-2}$ is between 1.96 and 1.99, so the interval is $-1.03 \pm 2(0.06)$ to two decimals, that is $(-1.15, -0.91)$; the choice between 1.96 and 2 does not change the printed answer.

```

1 b1 <- -1.03; se1 <- 0.06
2 c(t_stat = b1/se1, p_value = 2*pt(-abs(b1/se1), df = 100),
3   lower_t = b1 - qt(0.975, 100)*se1, upper_t = b1 + qt(0.975, 100)*se1,
4   lower_z = b1 - qnorm(0.975)*se1, upper_z = b1 + qnorm(0.975)*se1,
5   n_for_1pct_gap = which(qt(0.975, (3:400) - 2) /
6                           qnorm(0.975) < 1.01)[1] + 2)

```

##	t_stat	p_value	lower_t	upper_t
##	-1.717e+01	1.412e-31	-1.149e+00	-9.110e-01
##	lower_z	upper_z	n_for_1pct_gap	
##	-1.148e+00	-9.124e-01	1.250e+02	

The t statistic is -17.2 , so the slope is overwhelmingly significant; the interval excludes zero by a wide margin, which is the same statement.

By the simple-regression variance corollary of Unit 2, $\text{SE}(b_1) = s/(s_x\sqrt{n-1})$, so the width scales as $n^{-1/2}$ at fixed s and s_x : halving it needs four times the data. Widening the range of x is the cheaper route, since $\text{SE}(b_1)$ falls like s_x^{-1} . ■

Exercise 3.11. A categorical variable has 5 levels. How many indicators enter the model? What does each coefficient mean, and what changes if the reference level changes?

Solution. Four indicators, one level being the reference (Definition 3.10); including all five with an intercept makes $\mathbf{X}$ singular (Proposition 3.11).

Each coefficient is the difference in $\mathbf{E}[y]$ between its level and the reference, at fixed values of the other explanatory variables. Changing the reference changes all four coefficients, their standard errors and their t statistics, each being a comparison with a different baseline. Fitted values, residuals, SSE, R^2 and the partial F for the factor are unchanged, the column space of $\mathbf{X}$ being the same. ■

Exercise 3.12. An alternative to reference coding constrains the c level effects to sum to zero, fitting $c - 1$ columns whose j -th is 1 at level j , -1 at level c and 0 elsewhere. Show this spans the same column space, so every fitted value is unchanged, and say what the coefficients then mean. Verify on the **rad** factor of §6.4.

Solution. Let $\mathbf{D}_1, \dots, \mathbf{D}_c$ be the level indicators, so $\sum_j \mathbf{D}_j = \mathbf{1}$. Reference coding uses $\{\mathbf{1}, \mathbf{D}_1, \dots, \mathbf{D}_{c-1}\}$; effect coding uses $\{\mathbf{1}, \mathbf{D}_1 - \mathbf{D}_c, \dots, \mathbf{D}_{c-1} - \mathbf{D}_c\}$. Each effect column is a combination of $\mathbf{D}_j$ and $\mathbf{D}_c = \mathbf{1} - \sum_{i < c} \mathbf{D}_i$, hence of the reference basis, and conversely. The two span the same space, so by the projection characterisation of least squares in Unit 2 the fitted values, residuals, SSE, R^2 and every F statistic are identical; only the coefficients differ, being coordinates in a different basis.

Under effect coding the mean at level $j < c$ is $\alpha + \gamma_j$ and at level c is $\alpha - \sum_{i < c} \gamma_i$, so the c implied effects sum to zero and α is the unweighted average of the c level means at fixed other predictors. Each γ_j is a deviation from that average.

```
1 m_eff <- lm(medv ~ lstat + C(rad_f, contr.sum), data = Boston)
2 c(max_fitted_diff = max(abs(fitted(m_fac) - fitted(m_eff))),
3   SSE_difference   = sum(residuals(m_fac)^2) - sum(residuals(m_eff)^2),
```

```

4 sum_of_printed_effects = sum(coef(m_eff)[- (1:2)]),
5 implied_last_level     = -sum(coef(m_eff)[- (1:2)]))

```

```

##          max_fitted_diff          SSE_difference
##          1.421e-14              -3.638e-12
## sum_of_printed_effects    implied_last_level
##          1.074e+00              -1.074e+00

```

The ninth level's effect is not reported and must be recovered by subtraction, which is the practical drawback. 

Exercise 3.13. `rad` is an integer index of highway accessibility. Fitting it as a number rather than a factor is a restriction; write it as $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, count the constraints, and test it.

Solution. With $c = 9$ observed levels $r_1 < \dots < r_9$, the factor model has a free shift γ_j for each level, 8 free parameters after the reference. The numeric model imposes $\gamma_j = \beta(r_j - r_1)$ for a single β , forcing the 8 shifts onto a line: $8 - 1 = 7$ constraints, each of the form $\gamma_j / (r_j - r_1) = \gamma_2 / (r_2 - r_1)$, linear in $\boldsymbol{\beta}$ and so expressible as 7 rows of $\mathbf{R}$ with $\mathbf{r} = \mathbf{0}$.

```

1 m_num <- lm(medv ~ lstat + rad, data = Boston)

```

```
2 anova(m_num, m_fac)
```

```
## Analysis of Variance Table
##
## Model 1: medv ~ lstat + rad
## Model 2: medv ~ lstat + rad_f
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      503 19447
## 2      496 18128  7      1320 5.16 0.000011 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The residual degrees of freedom confirm the count, 503 against 496. The linearity restriction is decisively rejected: the effect of **rad** on **medv** is not proportional to the index, which is unsurprising for an ordering rather than a measurement. Coding an ordinal variable as a number is a testable assumption, and this is the test. 

Exercise 3.14. In $E[y] = 3 + 0.8x + 2z - 0.5xz$ with z binary, give the slope in x for each group and the difference in $E[y]$ between groups at $x = 0$ and $x = 6$. At which x do the two groups have the same mean?

Solution. Slope 0.8 when $z = 0$ and $0.8 - 0.5 = 0.3$ when $z = 1$.

The group difference is $E[y \mid z = 1] - E[y \mid z = 0] = 2 - 0.5x$: it is 2 at $x = 0$ and -1 at $x = 6$. With an interaction present the group difference is not a single number, and reporting the coefficient 2 as “the effect of z ” is correct only at $x = 0$.

The lines cross where $2 - 0.5x = 0$, at $x = 4$. Which group is ahead depends on where in the x range one looks. If $x = 0$ lies outside the observed range, the coefficient 2 describes a comparison that never occurs in the data. ■

Exercise 3.15. Fit `medv ~ lstat * chas`, then refit with `lstat` centred. Predict which coefficients move, check, and say which `chas` coefficient to report.

Solution. Centring replaces x by $x - \bar{x}$, so $\beta_0 + \beta_1 x + \beta_2 z + \beta_3 xz$ becomes $(\beta_0 + \beta_1 \bar{x}) + \beta_1(x - \bar{x}) + (\beta_2 + \beta_3 \bar{x})z + \beta_3(x - \bar{x})z$: the slope β_1 and the interaction β_3 are unchanged, while the intercept and the z coefficient each shift by $\bar{x}$ times the coefficient they pair with. The column space is unchanged, so every fitted value is identical.

```
1 Bc3 <- transform(Boston, lstat_c = lstat - mean(lstat))
2 mic <- lm(medv ~ lstat_c*chas, data = Bc3)
```

```
3 rbind(raw = coef(m_int), centred = coef(mic))
```

```
##           (Intercept)  lstat  chas lstat:chas
## raw           33.77 -0.915  9.825    -0.4329
## centred       22.19 -0.915  4.348    -0.4329
```

```
1 c(max_fitted_diff = max(abs(fitted(m_int) - fitted(mic))),
2   chas_at_mean_lstat = coef(m_int)[["chas"]] +
3   coef(m_int)[["lstat:chas"]]*mean(Boston$lstat),
4   t_chas_raw = coef(summary(m_int))["chas", 3],
5   t_chas_centred = coef(summary(mic))["chas", 3])
```

```
##      max_fitted_diff  chas_at_mean_lstat      t_chas_raw
##           3.553e-14           4.348e+00           4.672e+00
##      t_chas_centred
##           4.012e+00
```

Report the centred one. The raw 9.83 is the fitted river premium at `lstat` = 0, a value the data do not contain (the minimum is 1.73), and its t statistic tests a hypothesis about that non-existent point. Neither number is wrong; they answer different questions, and only one is about the data.



Exercise 3.16 (ISLR §3.7, exercise 3). A model for starting salary in thousands has $X_1 = \text{GPA}$, $X_2 = \text{IQ}$, $X_3 = \text{Level}$ (1 for College, 0 for High School), $X_4 = X_1X_2$ and $X_5 = X_1X_3$, with fitted coefficients $\hat{\beta}_0 = 50$, $\hat{\beta}_1 = 20$, $\hat{\beta}_2 = 0.07$, $\hat{\beta}_3 = 35$, $\hat{\beta}_4 = 0.01$, $\hat{\beta}_5 = -10$. (a) At fixed IQ and GPA, who earns more, and under what condition? (b) Predict the salary of a college graduate with IQ 110 and GPA 4.0. (c) True or false: since $\hat{\beta}_4$ is very small there is little evidence of a GPA/IQ interaction.

Solution. (a) The college premium at fixed GPA and IQ is $\hat{\beta}_3 + \hat{\beta}_5 \text{ GPA} = 35 - 10 \text{ GPA}$, which is positive for $\text{GPA} < 3.5$ and negative above. So college graduates earn more at low GPA and high school graduates earn more once GPA exceeds 3.5. This is Exercise 3.14 again: with an interaction the group difference is a function, not a number, and neither “college earns more” nor “high school earns more” is a complete answer.

(b) $50 + 20(4.0) + 0.07(110) + 35(1) + 0.01(4.0)(110) - 10(4.0)(1) = 50 + 80 + 7.7 + 35 + 4.4 - 40 = 137.1$ thousand.

(c) False, for two reasons. First, evidence about a coefficient is $|t| = |\hat{\beta}_4|/\text{SE}(\hat{\beta}_4)$, and no standard error is given: a small coefficient with a very small standard error is strong evidence. Second, a

coefficient's size depends on the units of its regressor. Here $X_4 = \text{GPA} \times \text{IQ}$ ranges to about $4 \times 130 = 520$, so the term $\hat{\beta}_4 X_4$ contributes about 5.2, comparable with the 7.7 contributed by IQ itself. By the invariance argument of Unit 2, rescaling X_4 divides $\hat{\beta}_4$ and its standard error by the same factor and leaves t alone, which is why t and not $\hat{\beta}$ is what one reads. ■

Exercise 3.17 (ISLR §3.7, exercise 10). Using `ISLR2::Carseats`, fit `Sales` on `Price`, `Urban` and `US`. (a) Interpret each coefficient, being careful with the qualitative ones, and write the fitted model as an equation. (b) For which predictors is $H_0: \beta_j = 0$ rejected? (c) Fit the smaller model those predictors suggest, compare the fits, and give 95% confidence intervals. (d) Is there evidence of high-leverage or outlying observations?

Solution.

```
1 CS <- ISLR2::Carseats
2 mfull <- lm(Sales ~ Price + Urban + US, data = CS)
3 round(coef(summary(mfull)), 4)
```

```
##           Estimate Std. Error  t value Pr(>|t|)
## (Intercept)  13.0435     0.6510  20.0357   0.0000
```

```
## Price      -0.0545      0.0052 -10.3892    0.0000
## UrbanYes   -0.0219      0.2717  -0.0807    0.9357
## USYes      1.2006      0.2590   4.6347    0.0000
```

```
1 msmall <- lm(Sales ~ Price + US, data = CS)
2 c(R2_full = summary(mfull)$r.squared, R2_small = summary(msmall)$r.squared,
3   adj_full = summary(mfull)$adj.r.squared,
4   adj_small = summary(msmall)$adj.r.squared)
```

```
## R2_full R2_small adj_full adj_small
## 0.2393  0.2393  0.2335  0.2354
```

```
1 round(confint(msmall), 4)
```

```
##           2.5 %  97.5 %
## (Intercept) 11.7903 14.2713
## Price      -0.0648 -0.0442
## USYes      0.6915  1.7078
```

```
1 c(max_leverage = max(hatvalues(msmall)), threshold = 2*3/nrow(CS),
2   n_above = sum(hatvalues(msmall) > 2*3/nrow(CS)),
3   max_abs_studentised = max(abs(rstudent(msmall))))
```

```
##          max_leverage          threshold          n_above
##          0.04334          0.01500          20.00000
## max_abs_studentised
##          2.89152
```

(a) **Urban** and **US** are two-level factors, so R codes each as one indicator (Definition 3.10) with reference levels **No**. The fitted model is

$$\widehat{\text{Sales}} = 13.043 - 0.0545 \text{Price} - 0.0219 \mathbb{1}_{\{\text{Urban}\}} + 1.201 \mathbb{1}_{\{\text{US}\}}.$$

A one-dollar rise in price is associated with 0.0545 thousand fewer unit sales at fixed location; a store in the US sells 1.201 thousand more than one outside it at the same price and urban status; the urban effect is -0.022 thousand, essentially nothing.

(b) **Price** and **US**, both with $p < 10^{-5}$. **UrbanYes** has $t = -0.08$ and $p = 0.94$.

(c) Dropping **Urban** changes R^2 in the fifth decimal and **raises** $\bar{R}^2$, exactly as Unit 2's $|t| > 1$ criterion predicts for a variable with $|t| = 0.08$. The intervals are $(-0.065, -0.044)$ for **Price** and $(0.692, 1.708)$ for **USYes**, neither containing zero.

(d) The maximum leverage is 0.0433 against a conventional threshold of $2(k+1)/n = 0.015$, so some points are flagged as high leverage, but none is near 1 and the design has no singleton (Unit

2's $h_{ii} = 1$ case). The largest studentised residual in absolute value is 2.89, which for $n = 400$ is unremarkable: the expected maximum of 400 independent $|N(0, 1)|$ variables is $\Phi^{-1}(1 - 1/802) = 3.02$. There is no outlier here in any useful sense. ■

Exercise 3.18. For $E[\ln y] = \beta_0 + \beta_1 x$: (a) give the exact percentage change in the fitted y per unit of x and its error against $100\beta_1$ at $\beta_1 = 0.05, 0.2, 0.5$; (b) show that if $\ln y = \beta_0 + \beta_1 x + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$ then $E[y \mid x] = e^{\beta_0 + \beta_1 x} e^{\sigma^2/2}$, so exponentiating a fitted log-mean estimates a median; (c) check (b) against a known truth.

Solution. (a) A one-unit increase multiplies the fitted y by e^{β_1} , a change of $100(e^{\beta_1} - 1)\%$. The approximation $100\beta_1$ is the first term of $e^u = 1 + u + u^2/2 + \dots$, so its relative error is about $\beta_1/2$.

```
1 bt <- c(0.05, 0.2, 0.5)
2 rbind(beta = bt, exact_pct = 100*(exp(bt) - 1), approx = 100*bt,
3       error = 100*(exp(bt) - 1) - 100*bt)
```

```
##           [,1] [,2] [,3]
## beta      0.0500 0.20 0.50
```

```
## exact_pct 5.1271 22.14 64.87
## approx    5.0000 20.00 50.00
## error      0.1271  2.14 14.87
```

(b) $y = e^{\beta_0 + \beta_1 x} e^\varepsilon$ and, for $\varepsilon \sim N(0, \sigma^2)$, the moment generating function gives $E[e^\varepsilon] = e^{\sigma^2/2}$, so $E[y | x] = e^{\beta_0 + \beta_1 x} e^{\sigma^2/2}$. The median of e^ε is 1 by symmetry, so the median of y is $e^{\beta_0 + \beta_1 x}$ exactly. Exponentiating the fitted log-mean therefore estimates the conditional median; the mean needs the factor $e^{\hat{\sigma}^2/2}$, or, without assuming normality, Duan's factor $n^{-1} \sum_i e^{e_i}$.

```
1 set.seed(41)
2 N <- 400000; x1 <- runif(N, 0, 10); sg <- 0.6
3 lg <- 1 + 0.2*x1 + rnorm(N, sd = sg); y1 <- exp(lg)
4 ml <- lm(lg ~ x1); s2l <- summary(ml)$sigma^2
5 p0 <- predict(ml, data.frame(x1 = 5))
6 sel <- abs(x1 - 5) < 0.02
7 c(true_mean = exp(1 + 0.2*5 + sg^2/2), true_median = exp(1 + 0.2*5),
8   sample_mean = mean(y1[sel]), sample_median = median(y1[sel]),
9   exp_fitted = exp(p0), times_smear = exp(p0)*exp(s2l/2),
10  duan = exp(p0)*mean(exp(residuals(ml))))
```

```
##      true_mean    true_median  sample_mean sample_median
```

##	8.846	7.389	8.941	7.533
##	exp_fitted.1	times_smear.1	duan.1	
##	7.402	8.864	8.862	

(c) The uncorrected value tracks the median and both corrections track the mean, agreeing with each other to 0.02%. The gap is 20%, which is the error made by anyone who fits in logs and reports $e^{\hat{y}}$ as an expected claim amount. ■

Exercise 3.19. State the Frisch–Waugh–Lovell theorem and use it to explain what “holding the other variables constant” means when no two observations share the same values of those variables. What does the theorem say when $\mathbf{X}_1$ is orthogonal to $\mathbf{X}_2$?

Solution. The statement and proof are Theorem 3.13: $\hat{\beta}_1$ from the full regression equals the coefficient from regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$, the residuals after projecting out $\mathbf{X}_2$.

So no observations need share values of $\mathbf{X}_2$. The procedure removes from both $\mathbf{Y}$ and $\mathbf{X}_1$ the part linearly predictable from $\mathbf{X}_2$ and relates what is left. “Holding constant” is an arithmetic adjustment for the **linear** influence of the measured variables, not a physical control.

If $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ then $\mathbf{M}_2\mathbf{X}_1 = \mathbf{X}_1 - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{X}_1 = \mathbf{X}_1$, so (9) reduces to $\hat{\beta}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y}$:

the coefficient is what it would have been had $\mathbf{X}_2$ never entered. Orthogonal predictors can be fitted one at a time with no change to any estimate, which is the point of an orthogonal experimental design and why sequential and marginal sums of squares coincide there and nowhere else. ■

Exercise 3.20. Continue Theorem 3.13 with $\mathbf{X}_1$ a single column. (a) The auxiliary regression reproduces $\hat{\beta}_1$ but not its reported t : explain and give the exact correction. (b) Show the auxiliary R^2 is the partial R^2 $(\text{SSE}_{-1} - \text{SSE})/\text{SSE}_{-1}$. (c) Check both on `lstat`.

Solution. (a) The two fits have the same coefficient and the same residual vector, hence the same SSE; what differs is the divisor. The auxiliary regression has one regressor and no intercept, so its reported s^2 is $\text{SSE}/(n-1)$, while the full model's is $\text{SSE}/(n-k-1)$. Every standard error, and so every t , is out by $\sqrt{(n-1)/(n-k-1)}$: multiply the auxiliary t by $\sqrt{(n-k-1)/(n-1)}$. The auxiliary fit has already spent $k-1$ degrees of freedom projecting out $\mathbf{X}_2$, and its degree-of-freedom count does not know it.

(b) Both regressions have residual sum of squares SSE, and the auxiliary total sum of squares is $\|\mathbf{M}_2\mathbf{Y}\|^2$, the residual sum of squares of $\mathbf{Y}$ on $\mathbf{X}_2$ alone, that is SSE_{-1} . So its R^2 is $1 - \text{SSE}/\text{SSE}_{-1}$. No centering correction is needed: $\mathbf{M}_2\mathbf{Y}$ has mean zero when $\mathbf{X}_2$ contains the intercept.

```

1 iaux <- lm(y_res ~ x_res - 1)
2  short2 <- lm(medv ~ age + rm, data = Boston)
3  c(coef_iaux = coef(iaux)[["x_res"]], coef_full = b[["lstat"]],
4    t_iaux = coef(summary(iaux))["x_res", 3],
5    t_corrected = coef(summary(iaux))["x_res", 3]*sqrt((n-k-1)/(n-1)),
6    t_full = coef(summary(fit))["lstat", 3],
7    R2_iaux = summary(iaux)$r.squared,
8    partial_R2 = (sum(residuals(short2)^2) - sum(residuals(fit)^2)) /
9                sum(residuals(short2)^2))

```

```

##      coef_iaux   coef_full      t_iaux t_corrected      t_full
##      -0.6685    -0.6685    -12.3352    -12.2985    -12.2985
##      R2_iaux   partial_R2
##      0.2315      0.2315

```

The partial R^2 is the right measure of what one predictor adds, and it is not the drop in the model's R^2 : the two differ by the factor $\text{SSE}_{-1}/\text{SST}$. ■

Exercise 3.21 (ISLR §3.7, exercise 15). For **Boston**, regress **crim** on each other variable one at a time, then on all of them together. Plot or tabulate the marginal coefficient against the partial

one for each predictor, and explain the pattern using Theorems 3.13 and 3.14.

Solution.

```
1 prd <- setdiff(names(Boston), c("crim", "rad_f"))
2 marg <- sapply(prd, function(v)
3   coef(lm(reformulate(v, "crim"), data = Boston))[[2]])
4 part <- coef(lm(crim ~ ., data = Boston[, c("crim", prd)]))[-1]
5 round(cbind(marginal = marg, partial = part[prd],
6             ratio = part[prd]/marg), 4)
```

##	marginal	partial	ratio
## zn	-0.0739	0.0457	-0.6182
## indus	0.5098	-0.0584	-0.1145
## chas	-1.8928	-0.8254	0.4361
## nox	31.2485	-9.9576	-0.3187
## rm	-2.6841	0.6289	-0.2343
## age	0.1078	-0.0008	-0.0079
## dis	-1.5509	-1.0122	0.6527
## rad	0.6179	0.6125	0.9912
## tax	0.0297	-0.0038	-0.1269

```
## ptratio    1.1520 -0.3041 -0.2640
## lstat      0.5488  0.1388  0.2529
## medv      -0.3632 -0.2201  0.6059
```

```
1 c(sign_flips = sum(sign(marg) != sign(part[prd])), predictors = length(prd),
2   correlation = cor(marg, part[prd]))
```

```
## sign_flips predictors correlation
##          7.000         12.000        -0.974
```

Every marginal coefficient is the omitted-variable expression of Theorem 3.14 with all eleven other predictors omitted: it carries the variable's own partial effect plus the part of every other variable's effect that travels with it. Every partial coefficient is the Frisch–Waugh–Lovell slope of Theorem 3.13, computed on what is left of both **crim** and the predictor after the other ten are projected out.

Seven of the twelve change sign, and **nox** is the extreme case: marginally 31.2, partially -10.0 . Nitrogen oxide concentration tracks the industrial, low-status, high-tax tracts where crime is high, and once those are in the model it carries no crime signal of its own and is negative. The two columns answer different questions, and neither is wrong: the marginal coefficient is the right

answer to “what do high-**nox** tracts look like”, the partial one to “what happens if **nox** moves and nothing else does”. Only the second is what a regression coefficient is usually taken to mean, and it is the one that depends on which other variables were measured. ■

Exercise 3.22. A regression of medical cost on age omits a health-status variable that raises cost and is positively correlated with age. What is the sign of the bias in the age coefficient?

Solution. By (10) with one included and one omitted variable the bias is $\delta\beta_2$, where δ is the slope from regressing health status on age and β_2 its effect on cost. Both are positive, so the bias is positive: the age coefficient overstates the association of age with cost, absorbing the part due to poorer health among older policyholders. ■

Exercise 3.23. Suppose the omitted $\mathbf{X}_2$ satisfies $\mathbf{X}'_1\mathbf{X}_2 = \mathbf{0}$ in the sample. (a) Show $\tilde{\beta}_1$ is not merely unbiased but numerically identical to the full-model $\hat{\beta}_1$, in every sample. (b) Show $\mathbf{E}[s^2_{\text{short}}]$ exceeds σ^2 and give the inflation. (c) Deduce what happens to standard errors and prediction intervals, and check by simulation with $\beta_2 = 3$, $\sigma = 1$. (d) Which of the following, from a standard list of consequences of omitting an important variable, remain true here: underfitting;

wider prediction intervals; unbiased coefficients; inflated s^2 ; unchanged SST?

Solution. (a) By Exercise 3.19, orthogonality makes $\mathbf{M}_2\mathbf{X}_1 = \mathbf{X}_1$, so the full-model coefficient is $(\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{Y}$, which is the short regression's estimator verbatim. This is stronger than unbiasedness: the two are the same function of the data.

(b) The short model's residual vector is $\mathbf{M}_1(\mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon})$ with $\mathbf{M}_1 = \mathbf{I} - \mathbf{X}_1(\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1$, and orthogonality gives $\mathbf{M}_1\mathbf{X}_2 = \mathbf{X}_2$, so by the quadratic-form lemma of Unit 2

$$\mathbb{E}[\text{SSE}_{\text{short}}] = \|\mathbf{X}_2\boldsymbol{\beta}_2\|^2 + \sigma^2(n - k_1 - 1), \quad \mathbb{E}[s_{\text{short}}^2] = \sigma^2 + \frac{\|\mathbf{X}_2\boldsymbol{\beta}_2\|^2}{n - k_1 - 1} :$$

the omitted systematic variation is counted as noise.

(c) Standard errors are $s\sqrt{[(\mathbf{X}'_1\mathbf{X}_1)^{-1}]_{jj}}$, so they inflate by $\sqrt{\mathbb{E}[s_{\text{short}}^2]/\sigma^2}$, and prediction intervals inflate at least as much. The coefficient is right and its uncertainty is overstated.

```

1 nn <- 200
2 z1 <- rep(c(-1, 1), length.out = nn)           # orthogonal by construction
3 z2 <- rep(c(-1, -1, 1, 1), length.out = nn)
4 c(sum_z1 = sum(z1), sum_z2 = sum(z2), inner = sum(z1*z2))

```

```
## sum_z1 sum_z2 inner
##      0      0      0

1 set.seed(42)
2 est3 <- t(replicate(4000, {
3   yo <- 1 + 2*z1 + 3*z2 + rnorm(nn)
4   c(full = coef(lm(yo ~ z1 + z2))["z1"]),
5     short = coef(lm(yo ~ z1))["z1"]))
6 })
7 yo <- 1 + 2*z1 + 3*z2 + rnorm(nn)
8 mfl <- lm(yo ~ z1 + z2); msh <- lm(yo ~ z1)
9 c(mean_full = mean(est3[, 1]), mean_short = mean(est3[, 2]),
10   sd_full = sd(est3[, 1]), sd_short = sd(est3[, 2]),
11   s_full = summary(mfl)$sigma, s_short = summary(msh)$sigma,
12   se_ratio = coef(summary(msh))["z1", 2]/coef(summary(mfl))["z1", 2],
13   predicted_ratio = sqrt(1 + 9*mean(z2^2)))
```

```
##      mean_full      mean_short      sd_full
##      2.00194      2.00194      0.06962
##      sd_short      s_full      s_short
##      0.06962      0.97999      3.07317
```

##	se_ratio	predicted_ratio
##	3.13592	3.16228

The two estimators agree to the last digit in mean and standard deviation, confirming (a) rather than merely (b). On one sample s rises from 0.98 to 3.07 and the standard error by a factor of 3.14 against the predicted $\sqrt{10} = 3.16$.

(d) Underfitting, wider prediction intervals, inflated s^2 and unchanged SST all hold: $\text{SST} = \sum_i (Y_i - \bar{Y})^2$ involves the response alone, so no change to $\mathbf{X}$ can move it. The one that fails in general is unbiasedness, by Theorem 3.14, and orthogonality is precisely the exception in which it survives. So omitting a relevant orthogonal variable costs precision without costing correctness; omitting a correlated one costs both. ■

Exercise 3.24 (SOA sample question 27). Monthly dental claims are modelled on 48 observations with three predictors, giving

	coefficient	standard error	t	p
Intercept	-45,765,767.76	20,441,816.55	-2.24	0.0303
Weekdays	513,280.76	233,143.23	2.20	0.0330
Weekend days	280,148.46	483,001.55	0.58	0.5649
Average members	38.64	6.42	6.01	0.0000

Determine which predictor should be dropped at the 5% level. Reproduce the t column and the p column from the first two, and say why the three tests should not be read as a single 5% decision.

Solution. Only Weekend days has $p > 0.05$, so it is the one to drop; the other two and the intercept are significant. Each t is the coefficient over its standard error (Theorem 3.1) and each p is $2\mathbf{P}\{t_{n-k-1} > |t|\}$ with $n - k - 1 = 48 - 3 - 1 = 44$.

```

1 cf27 <- c(-45765767.76, 513280.76, 280148.46, 38.64)
2 se27 <- c(20441816.55, 233143.23, 483001.55, 6.42)
3 tv27 <- cf27/se27
4 rbind(t = round(tv27, 2), p = round(2*pt(-abs(tv27), df = 44), 4),
5       drop_at_5pct = as.numeric(2*pt(-abs(tv27), df = 44) > 0.05))

```

```
##           [,1]  [,2]   [,3] [,4]
## t        -2.2400 2.200 0.5800 6.02
## p         0.0303 0.033 0.5649 0.00
## drop_at_5pct 0.0000 0.000 1.0000 0.00

1  c(familywise_3_tests = 1 - 0.95^3, bonferroni_level = 0.05/3,
2    weekend_sig_at_bonferroni =
3    as.numeric(2*pt(-abs(tv27[3]), df = 44) < 0.05/3))

##          familywise_3_tests          bonferroni_level
##                   0.14263                   0.01667
## weekend_sig_at_bonferroni
##                   0.00000
```

Three tests at 5% have a family-wise error rate of $1 - 0.95^3 = 0.143$ under the null, so the probability of dropping at least one useful predictor by chance is nearly three times the nominal level. Exercise 3.6 shows the same effect for confidence regions. Two further cautions: the predictors are dropped **one at a time**, since removing Weekend days changes every remaining coefficient and standard error unless the design is orthogonal (Exercise 3.23); and a joint test of the two weekday variables by Corollary 3.5 answers a different and often more relevant question than two

separate t tests, as Exercise 3.5 shows. ■

Exercise 3.25. A coefficient is significant at the 0.1% level with $n = 2,000,000$. What follows about its practical importance? What would you ask to see instead?

Solution. Nothing. $t = \hat{\beta}_j / \text{SE}(\hat{\beta}_j)$ and $\text{SE}(\hat{\beta}_j) \propto n^{-1/2}$, so at fixed effect size $|t|$ grows without bound in n : at two million observations an economically negligible coefficient is overwhelmingly significant.

Ask for the coefficient in the units of the response, multiplied by a change in the predictor that is meaningful in the application, for instance an interquartile shift, so the answer reads as an amount and not as a slope; the confidence interval for that amount; and, if the model is for prediction, the prediction interval, which at this n is essentially $\pm 2\sigma$ however precise the coefficients are (Exercise 3.9). ■

Exercise 3.26. On the fit of §6: (a) confirm that `anova` and `linearHypothesis` agree for a variable-exclusion hypothesis, and construct the reduced model that lets `anova` test $\beta_{\text{lstat}} = \beta_{\text{age}}$ too; (b) verify (4) against the reduced fit; (c) explain why the reduced fit's standard errors are not

the restricted standard errors of Exercise 3.2.

Solution.

```
1 pooled <- lm(medv ~ I(lstat + age) + rm, data = Boston)
2 c(anova_F = anova(pooled, fit)$F[2],
3   linearHyp_F = linearHypothesis(fit, "lstat = age")$F[2],
4   npar_pooled = length(coef(pooled)), npar_full = length(coef(fit)),
5   s2_full = s2, s2_reduced = sum(residuals(red)^2)/(n - 2))
```

```
##      anova_F linearHyp_F npar_pooled  npar_full      s2_full
##      120.69      120.69         3.00         4.00      30.72
## s2_reduced
##      38.64
```

(a) §6.2 runs the first comparison, with all four routes agreeing. `anova` needs the reduced model as a fitted object, which for $\beta_{\text{lstat}} = \beta_{\text{age}}$ means building the pooled regressor `I(lstat + age)`; it then reports the same F as `linearHypothesis`. The obstacle was never the hypothesis but that `anova` compares fitted objects.

(b) Exercise 3.2 does this: (4) returns `lm(medv ~ lstat)`'s coefficients with two exact zeros.

(c) The two use different estimates of σ^2 . The restricted standard errors evaluate $\sigma^2 \mathbf{P} \mathbf{A}^{-1} \mathbf{P}'$ at the

full model's $s^2 = 30.72$; the reduced fit estimates σ^2 from its own residuals, $\text{SSE}_R/(n-2) = 38.64$. Neither is wrong. Under H_0 both are consistent and the full model's has more degrees of freedom; if H_0 is false the reduced model's s^2 is inflated by the constraint, exactly as in Exercise 3.23(b), and its standard errors are too large. Which to report depends on whether the constraint is assumed or tested. ■